LESSON PLAN

EJECTION SEAT (36)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic requirements of a fission weapons
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic requirements of a fission weapons. Critical mass of fissile material. Basic working principles of fission weapons
5. Specific Objectives	36.1.1.1 Trainee will be able to explain conventional and nuclear explosion

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical concepts via Q&A; No hands-on practical activity available for fission weapon internals.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between a conventional and a nuclear explosion?
(b) Can you list two fissile materials used in nuclear weapons?
(c) What is the general purpose of the implosion mechanism in a fission weapon?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Conventional vs Nuclear Explosion • Fissile Material Requirements • Critical Mass Definition • Factors Affecting Critical Mass • Basic Fission Chain Reaction	L & Ds	PPT, IB	15 min
(b)	• Gun-type Assembly Mechanism • Implosion-type Assembly Mechanism • Neutron Initiator Function • Reflector Role in Criticality • Safety Mechanisms in Fission Weapons	L & Ds	PPT, PDF	20 min
(c)	• Summary of Fission Weapon Principles • Comparison of Assembly Types • Role of Plutonium-239 vs Uranium-235 • Importance of Neutron Economy • Transition to Thermonuclear Weapons	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between the gun-type and implosion-type assembly methods.
(b) Why is achieving critical mass essential for a nuclear explosion?
(c) How does a neutron initiator contribute to the fission chain reaction?

12. Home Assignment and Follow Up	Prepare a brief note comparing the fissile material requirements for gun-type versus implosion-type fission weapons, referencing concepts from the previous lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to thermo nuclear weapons. Fission – fusion and fission –fusion –fission process	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min